



Loi Chong Bin

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Portfolio: <http://bit.ly/4tKPILR>

Bridge design engineer with three years of experience in structural analysis, project coordination, and technical design of reinforced concrete and prestressed bridges. Proven record in cost-effective, standards-compliant infrastructure design. Demonstrated capabilities in coordinating with multidisciplinary teams in the successful delivery of structural engineering projects. Proficient in BS5400 and Eurocode bridge design, with BEM Graduate Engineer registration with a strong interest in applying RC and prestressed design expertise toward innovative and sustainable construction technologies.

WORK EXPERIENCE

OSD CONSULTANTS

FEB 2023 - PRESENT

Structural Design Engineer

Ara Damansara, PJ

- Optimise the design of reinforced concrete and prestressed concrete bridges across multiple projects, delivering cost-effective and construction-friendly designs.
- Model reinforced and prestressed concrete bridges using STRAP and MIDAS Civil, and conducting structural analysis, ensuring design compliance with British Standards and Eurocode.
- Identified design-construction discrepancies during site inspections and proposed practical solutions, reducing on-site rework across all construction works.
- Assess and review shop drawings, work method statement, and RFI, ensuring compliance with industry standards, while maintaining project quality and safety.
- Deliver quantity take-offs and reinforcement poundage calculations to support accurate project cost estimations.

OSD CONSULTANTS

DEC 2021 - MAR 2022

Bridge Design Engineer (Intern)

Ara Damansara, PJ

- Supported senior engineers by collaborating on the development of design calculations, drawings and plans for reinforced concrete bridge projects, ensuring timely completion and adherence to engineering standards.
- Conducted structural analysis of various bridges using STRAP and ADSEC, providing results that informed the design of multiple bridges while ensuring structural integrity and compliance with engineering standards.

PROJECTS

GRANDHILL GENTING INFRA BRIDGE

JUL 2025 - PRESENT

- Responsible for the comprehensive design and modelling of the bridge's substructure and superstructure using STRAP, ensuring compliance with engineering standards and delivering a well-integrated RC design.
- Execute the design of Post-Tensioned T-Beam as the girders of the approach span, ensuring compliance with load requirements.
- Coordinate with engineers and drafters in producing detailed design drawings, ensuring timely completion of the project.
- Coordinate with contractors to clarify design intent and resolve technical issues, improving on-site execution.
- Deliver the design of RCC counterfort wall while ensuring structural stability and effective soil retention.

PAN BORNEO HIGHWAY

APR 2025 – PRESENT

- Lead the design of RC bridges for package WP32, WP33 and WP35.
- Delivered cost-efficient design optimizations resulting in estimated 20% construction cost savings.
- Conducted detailed reinforcement poundage calculations to support cost estimation and material planning.
- Coordinate with drafters in producing detailed design drawings, ensuring timely completion of the project.
- Execute the design of Post Tensioned LS Girders, ensuring compliance with load requirements.

JALAN YEW TO SG. BESI WIDENING

JAN 2024 – PRESENT

- Work closely with senior engineers and the primary design consultant to develop plans for widening of the existing Jalan Yew to Sungai Besi flyover.
- Collaborate with senior engineers to develop the Post-Tensioned FlexC Girder design, using analytical engineering to ensure structural integrity.
- Review shop drawing and coordinate with contractors, ensuring accurate compliance with specifications.
- Conduct impact assessment of the existing Jalan Yew to Sungai Besi flyover to ensure stability and safety during beam launching operations.
- Review RFI's raised by contractor, by providing feasible and technically sound solutions to resolve design and construction issues.

EDUCATION

MONASH UNIVERSITY

MAR 2019 – DEC 2022

BEng, Civil Engineering (Honours)

- Second Upper Class Honours (H2A)

Final Year Project: Flood Vulnerability Index (FVI) for Critical Infrastructure in Mixed-Use Development

- Applied qualitative and quantitative research methodologies, including focus group sessions, to evaluate flood vulnerability indexes and inform evidence-based findings.
- Created a flood vulnerability map using ArcGIS on which critical infrastructures are susceptible to flood, using the flood vulnerability equation

CERTIFICATIONS

NIOSH+

DEC 2025

- Expressway Operations Safety Passport

BOARD OF ENGINEERS MALAYSIA (BEM)

MAY 2023

- Graduate Engineer

CIDB HOLDINGS

FEB 2023

- Safety Induction Construction Workers (SICW)

SKILLS

MODELLING SOFTWARE - STRAP, ADSEC, Autodesk Structural Bridge Design, BEAMD, SAFE, CROSSEC, MIDAS, Tekla Tedds, SPACEGASS, ETABS, Prota, LUSAS

MICROSOFT OFFICE - Visio, Excel, PowerPoint, Word

CAD SOFTWARE - AutoCAD, ZWCAD

3D MODELLING SOFTWARE - SketchUp, Revit

GIS MODELLING SOFTWARE - ArcGIS

TRAFFIC MODELLING SOFTWARE – SIDRA

LANGUAGES

English (Native), Chinese (Native), Malay (Native)

LEADERSHIP

MONASH ATHLETICS CLUB

JAN 2022 – OCT 2022

Secretary

- Managed secretarial duties for the club, including meeting coordination, minutes documentation, and internal communications to ensure smooth organisational operations.
- Coached and mentored athletes in participating in the Kawan Athletics Competition by providing ample support and guidance to enhance performance and build team confidence.